

lockfix (runner patch) – guide

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Source: `constitution/scripts/codegraph/runner_patches/lockfix1.py` (73 lines). A **module driven by** `fk_index_patch.py`, not a command. Citations are `lockfix1.py:<line>` unless another file is named. Machinery guide: [fk_index_patch](#).

FINDING FIRST: this patch has no behavioural test

No file under `scripts/codegraph/tests/` exercises lockfix1's runtime behaviour (no test contains `__helixLockOwnerIsCodegraph`, and none builds a runner with lockfix1 and inspects a lock). The tree mentions the id only in selection arguments (`tests/test_bulk_classify_mutate.py`, `tests/test_mcp_readonly_mutate.py`, `tests/test_unit_safe.sh`) and `codegraph_safe.sh` (lines 63-68) states it has zero tests, which is why the wrapper does not use it by default (`--patches fkidx1,resolve1`). `owed_report.py` reports "tested" only because the id is named. Everything below is traced from source and from the ad hoc scratch checks recorded under Usage; nothing was run against the real runner.

Overview

Stock CodeGraph 1.6.0 `FileLock.acquire` (`lib/dist/utils.js`) treats a `.codegraph/codegraph.lock` whose mtime is older than `STALE_TIMEOUT_MS` (2 minutes) as stale **whatever the recorded PID is** and unlinks it, and nothing

refreshes the mtime (docstring lines 3-9). A second `sync / index / init / serve` started more than two minutes into a long index therefore deletes the live lock and writes the database concurrently.

lockfix1 removes the age clause. A lock is honoured only if its PID is an integer > 1 , the process is alive (vendor `isProcessAlive` , `kill(pid, 0)`) **and** `/proc/<pid>/cmdline` contains `codegraph` ; a recycled PID that now belongs to an unrelated program is treated as stale (lines 11-16, 36-38, 40-56).

A rejected alternative is recorded in the source (lines 18-22): refreshing the lock mtime from a timer, because the vendor's own comments say `parse/store/resolve` spans block the event loop for minutes.

Prerequisites

Reads `lib/dist/utils.js` from the package. Python stdlib plus `runner_patches/common.py` . Default-set patch (`runner_patches/___init___py:17`), applied second.

Usage

Only through `fk_index_patch.py` (see the parent guide for flags):

```
python3 scripts/codegraph/fk_index_patch.py --patches lockfix1 --dst <scratch dir>
```

`NOT RUN` in this form (copies the 282 MB package). Checks that were run, all in a scratch directory holding only a copy of stock 1.6.0 `utils.js` :

- VERIFIED (ran: `lockfix1.plan(FileView(scratch))`); saw: `APPLIED`, anchors `'class FileLock {'` and `'FileLock.acquire age-override condition'`, detail `owner_identity_check "/proc/<pid>/cmdline contains 'codegraph'"`) .
- VERIFIED (ran: `node --check` on the planned `utils.js` with the system `node v24.18.0`; saw: `exit 0`; control: `node --check` on a deliberately broken file exits 1 with `SyntaxError`) .

- VERIFIED (ran: the injected helper extracted from the planned text and run under node; saw, for the process's own PID with "codegraph" in its argv: true; without it: false; PID 1: false; PID 999999 (no such process): false; the string '5': false) . This confirms the identity test and the `pid <= 1` / non-integer refusals (helper lines 42-55).

Behaviour (source-traced)

`plan(view)` (lines 59-73):

1. If the age-based condition anchor (line 36) is absent **and** `class FileLock` is present, `STALE_TIMEOUT_MS` is gone and `isProcessAlive(pid)` is present, the patch reports NOT_NEEDED (lines 61-65).
2. Otherwise, if the helper marker `__helixLockOwnerIsCodegraph` is present, Refuse: `source is ALREADY patched` (lines 66-67).
3. `replace_once` on `class FileLock {` : inserts the helper function before the class (line 68).
4. `replace_once` on `if (lockAge < FileLock.STALE_TIMEOUT_MS && ! isNaN(pid) && this.isProcessAlive(pid)) {` : replaced by `if (! isNaN(pid) && pid > 1 && this.isProcessAlive(pid) && __helixLockOwnerIsCodegraph(pid)) {` (lines 36-38, 69).

Helper semantics (lines 40-56):

Situation	Helper returns
Not an integer, or <code><= 1</code>	<code>false</code> (lock is stale)
<code>/proc/self/cmdline</code> absent (no <code>procfs</code>)	<code>true</code> — owner cannot be disproven, never steal the lock
<code>/proc/<pid>/cmdline</code> unreadable with <code>ENOENT</code>	<code>false</code> — owner exited
Unreadable with any other error	<code>true</code> — keep the lock
Readable	<code>true</code> iff the command line contains <code>codegraph</code>

Honest boundary stated in source (lines 24-27): the lock file stores only a PID, so a lock written by a codegraph on another host over a shared filesystem is judged against the local PID table, exactly like stock; only the age override is removed.

Observed limitation (from the helper text, not run): the identity test is a substring match on `codegraph` anywhere in the command line, so any live process with that word in its arguments (an editor opened on a codegraph path, for example) keeps a lock alive. The check is deliberately biased towards never stealing.

Edge cases

All `VERIFIED` (ran: planning against a scratch copy of stock 1.6.0 `utils.js`) :

Condition	Result
Stock 1.6.0	APPLIED (2 anchors)
Plan again on the already-patched text	Refuse: lockfix1: source is ALREADY patched – --src must be the pristine package
<code>STALE_TIMEOUT_MS</code> and the age clause gone (upstream fixed)	NOT_NEEDED: <code>utils.js</code> <code>FileLock</code> no longer has an age-based stale override (<code>STALE_TIMEOUT_MS</code> gone, PID liveness still checked)
<code>class FileLock {</code> re-named	Refuse: lockfix1: anchor ' <code>class FileLock {</code> ' occurs 0x in <code>lib/dist/utils.js</code> (expected exactly 1) – upstream shape changed, re-derive the patch

Interaction: `mcpro1` also edits `utils.js` (`FileLock.acquire / release`) and is exclusive with every other patch, so the two can never be combined (`runner_patches/__init__.py:31-34`).

Related scripts

[fk_index_patch](#), siblings [fkidx1](#), [datafrag1](#), [resolve1](#), [resolve2](#), [mcpro1](#) (which removes lock acquisition entirely), [codegraph_safe](#) (declines it by default until tests exist), [owed_report](#).

Last verified

2026-09-26, scratch copy of stock 1.6.0 `utils.js` only; the real runner cache and every `.codegraph` were untouched.